

AI-Optimized Science Metadata Formats

Does representation matter when an LLM searches and reads CMR metadata?

Faceted vs. raw UMM × six serializations × three models, measured end to end

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The pitch we made

*The best representation is **not universal** — it is consumption-mode- and model-dependent, and must be measured empirically.*

Made falsifiable by crossing **format** against **content scope** — the confound the field has been ignoring.

H1 Content dominates

field selection matters more than format for retrieval.

H3 The axes disagree

retrieval, reasoning, and cost need not rank formats alike.

H2 Format effect is small

at equal content, representations barely differ.

H4 Best format varies

the top reasoning representation differs by model family.

Verdict up front

H1 Content dominates	Supported	Payload effect is the largest in the study, on every axis, for every model.
H2 Format effect is small	Supported*	At equal content, the largest format penalty is about half the payload effect. *JSON-LD is a real retrieval outlier.
H3 The axes disagree	Supported	JSON wins raw-record reasoning but sits 5th of 6 in retrieval; JSON-LD is last in retrieval yet mid-pack in reasoning.
H4 Best format varies	Suggestive	Three models, three different curated-payload winners — but within-model spreads overlap their confidence intervals.

Every verdict below comes from one audited pipeline: 10,800 judged answers over real CMR records.

Experimental setup — the grid

Three independent axes

- **Payload** — **faceted**: curated 32-field projection vs. **unfaceted**: raw UMM record (45 top-level fields, 398 paths)
- **Format** — JSON, YAML, TOON, CSV, JSON-LD, Metadata-as-Text (prose); all six carry identical facts (parity hard-gated at render)
- **Model** — gpt-5.4-nano, gpt-5.4-mini (cloud); muse-glimmer:30b (local 32B)

Held fixed — encoder (bge-large-en-v1.5), 510-token chunks, max pooling, $k=10$ records in context, temperature 0, seed, one judge

Scale

- 2,584 real CMR collection records
× 12 renderings
- 511 queries for retrieval (11 SME + 500 synthetic)
- 300-query slice for LLM stages
- $6 \times 2 \times 3 = 36$ arms
× 300 queries = **10,800 judged cells**
- Spend: \$71 OpenAI + ~145 h local generation

Experimental setup — staged pipeline, every number auditable

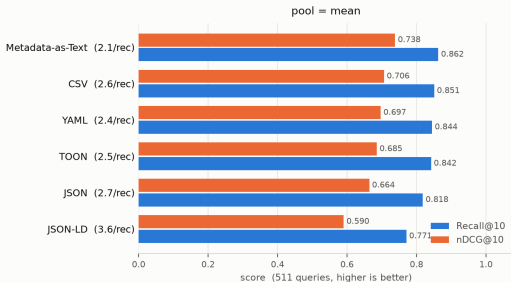
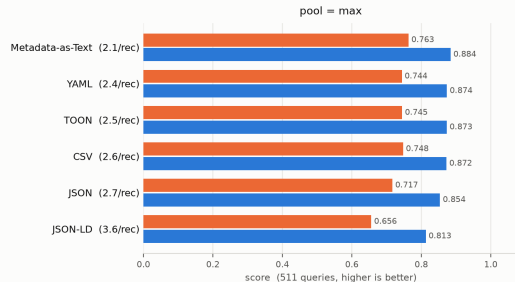
Stage	What it does	Measures
R Retrieve	chunked vector search per format/payload	Recall@k, MRR, nDCG@10
A Answer	top-10 records into each model, temp 0	prompt/completion tokens, latency, \$
J Judge	DeepEval metrics on frozen answers	correctness, faithfulness, ...

- Stages are frozen artifacts — judging re-runs never touch answers; checkpointed and resumable
- Correctness is the end-to-end metric (knows ground truth); the rest are diagnostics
- Content parity across formats is enforced, so format effects are representation-only

The same 60% token gap shows up before the LLM ever runs: faceted renders at 786–1,465 bge tokens/record; unfaceted at 1,462–7,012.

Stage R — retrieval: prose wins, JSON-LD loses, faceted leads

Chunked retrieval over faceted records — BAAI/bge-large-en-v1.5, 510-token chunks

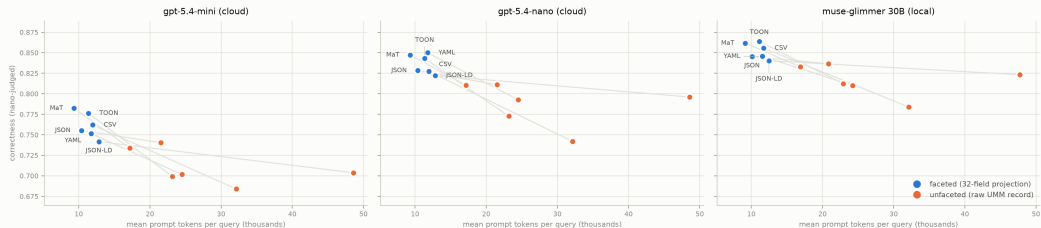


JSON-LD produces 1.7x more chunks per record than Metadata-as-Text for identical content (JSON-LD 3.6, JSON 2.7, CSV 2.6, TOON 2.5, YAML 2.4, Metadata-as-Text 2.1). Under max-pooling more chunks means more independent chances to match a query, so these scores reward verbosity. Read them as chunked retrieval over full records, not as a format ranking.

- **Metadata-as-Text is first in all four conditions** (both payloads $\times$ both poolings), on all five metrics — and it is also the **cheapest** format to embed
- **JSON-LD is last or near-last everywhere**, at 1.7–4.4 $\times$ the chunks of prose
- Faceted $\geq$ unfaceted for **every** format: $\Delta R@10$ +0.01 (prose) to +0.14 (CSV) — with a query-provenance caveat, later

Stage A/J — correctness vs. context cost (300 queries, all arms)

Answer correctness vs context cost — 300 queries, top-10 records in context

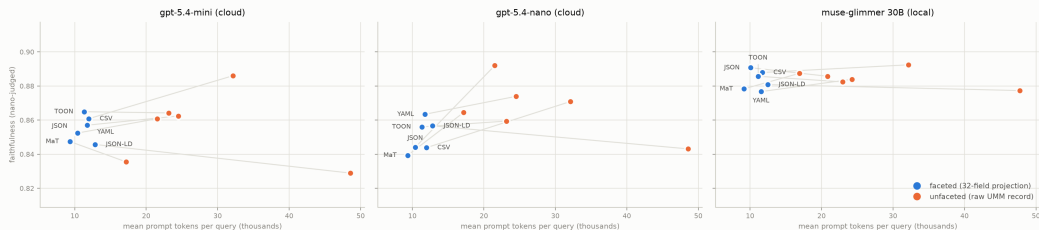


Labels name the faceted point; the gray connector leads to the same format's unfaceted point. Judge: gpt-5.4-nano; correctness pooled over 11 SME + 289 synthetic queries.

- **Faceted** sits up and to the left of **unfaceted** in every panel: **more correct at ~40% of the tokens**
- Payload effect +0.045 correctness [+0.038, +0.052], $n=5,400$ pairs; holds per model
- Cheapest strong point: faceted MaT $\approx 9.1\text{--}9.3\text{k}$ tokens/query, top-two correctness for all models

Stage A/J — faithfulness: no trade-off worth making

Answer faithfulness vs context cost — 300 queries, top-10 records in context



Labels name the faceted point; the gray connector leads to the same format's unfaceted point. Judge: gpt-5.4-nano; faithfulness pooled over 11 SME + 289 synthetic queries.

- Raw records buy -0.006 $[-0.012, -0.001]$ faithfulness back — barely detectable, at $2.5\times$ the tokens and -0.045 correctness
- The local model is the most grounded arm: glimmer $+0.025$ faithfulness over nano
- Faithfulness never separates the formats the way correctness separates the payloads

The payload axis is the largest effect in the study, on all three measurement axes.

Axis	Faceted vs. unfaceted	Largest format effect	Ratio
Reasoning (correctness)	+0.045 [+0.038, +0.052]	CSV −0.027 vs. JSON	~1.7×
Retrieval (R@10)	+0.01 to +0.14, every format	JSON-LD −0.07 faceted	similar [†]
Cost (prompt tokens)	−16,533/query (−60%)	~−3,500 within payload	~5×

- Per model: nano +0.049, mini +0.051, glimmer +0.036 — all clear of zero at $n=1,800$
- The curated projection is **better and cheaper for every model tested** — the run's headline finding, and it is not close
- [†]Retrieval caveat: the synthetic query generator saw a field summary that structurally resembles the faceted projection — payload gap there is an observation, not a finding

H2 — Format effect is small

Supported, with one outlier

At equal content, serialization moves correctness by at most half the payload effect.

Reasoning, paired vs. JSON ($n=1,800$ each)

MaT (prose)	+0.009	$[-0.003, +0.020]$
TOON	-0.008	$[-0.019, +0.003]$
YAML	-0.011	$[-0.022, -0.000]$
JSON-LD	-0.015	$[-0.027, -0.003]$
CSV	-0.027	$[-0.039, -0.015]$

- Only CSV and JSON-LD clearly separate from JSON — and by less than the payload's 0.045
- **Format matters far less than what you put in it**
- The outlier: at *retrieval*, JSON-LD trails by 0.07–0.10 R@10 while costing the most tokens — format is not *free* at the embedding stage

H3 — The axes disagree

Supported

No single format wins retrieval, reasoning, and cost at once — rankings genuinely diverge.

Format	Retrieval (chunked)	Reasoning, faceted	Reasoning, raw	Cost
MaT (prose)	1st, all 4 conditions	top-two, all models	mid	cheapest
JSON	5th of 6 (faceted)	mid	1st, all 3 models	mid
CSV	4th faceted / last raw	worst pooled penalty	last (2 of 3)	high (raw)
JSON-LD	last everywhere	mid-pack	mid	highest

- Even the judge's diagnostics disagree: contextual recall runs *counter* to correctness (faceted MaT 0.36 vs. raw JSON-LD 0.60 — while beating it 0.847 to 0.796 on correctness)
- Consequence: a repository cannot pick one representation off a single benchmark axis — exactly the claim in the pitch

H4 — Best format varies by model

Suggestive

Three models, three different winners under the curated payload — but the gaps are inside the error bars.

Model	Faceted winner	Raw-record winner
gpt-5.4-nano	YAML (0.850)	JSON (0.811)
gpt-5.4-mini	MaT (0.782)	JSON (0.740)
muse-glimmer:30b	TOON (0.864)	JSON (0.837)

- Under the **raw record**, **JSON wins for all three** — explicit structure appears to help when there are 398 paths to navigate; content scope *modulates* format sensitivity
- Within-model spreads (0.028–0.040) overlap the paired CIs (± 0.011 –0.012): the *ordering* is suggestive, not established — H4 needs more SME-hard queries to close
- What *is* established: no format is safe to hard-code across models

The model axis — interesting, and least trustworthy

Paired correctness ($n=3,600$ each)

mini – nano	−0.076
glimmer – nano	+0.022
mini – glimmer	−0.098

Read literally: a local 32B beats both cloud models, and the small cloud model beats the larger one.

Before believing that:

- The judge **is** gpt-5.4-nano — it grades its own answers in two of three comparisons, and wins both. Self-preference cannot be separated from quality here
- The one **judge-neutral** comparison (mini vs. glimmer) is the *largest* effect — and glimmer's win over nano is biased *against* glimmer, so it is conservative
- Slice size matters: glimmer's edge read +0.067 at 50 queries, +0.022 at 300 — the small slice was SME-heavy, glimmer's best terrain

The hard part: expert questions

SME-written queries score ~ 0.3 – 0.4 correctness below synthetic ones, for every model.

Model (faceted, range over formats)	SME ($n=11$)	Synthetic ($n=289$)	Gap
gpt-5.4-nano	0.36–0.48	0.84–0.87	~ 0.43
gpt-5.4-mini	0.28–0.42	0.76–0.80	~ 0.40
muse-glimmer:30b	0.47–0.63	0.85–0.87	~ 0.31

- Synthetic queries inherit vocabulary from the records they were generated from; expert questions do not — **the SME slice is the realistic difficulty**
- Pooled means are dominated by slice composition; compare per-slice, never pooled
- The local model has the smallest expert gap and the best expert scores — worth chasing, but $n=11$ is far too few expert queries to conclude anything

What we would push back on ourselves

- **The judge is a model under test.** All quality scores are nano-graded; the model ranking is the result we trust least. (The format and payload effects are within-judge comparisons and far more robust.)
- **Query provenance leans faceted, and $\sim 3/4$ of the payload effect is retrieval-mediated.** The synthetic generator consumed a curated field summary, and each payload retrieves its own contexts. With retrieval equalized (both payloads at $R@10 = 1$), faceted still wins — but by $+0.012$, not $+0.045$.
- **One encoder, one chunking scheme.** Prose ranked *last* under a different encoder on whole documents and first here — retrieval rankings are encoder-conditional. Chunk boundaries also cut JSON mid-object but prose mid-paragraph.
- **Format rankings within a model are not statistically separated** — only the pooled CSV and JSON-LD penalties clear their intervals.
- **Expert coverage is thin:** 11 SME queries carry all realistic difficulty.

Future work

Close the loops this run opened

- **Neutral judge re-run** — re-grade a slice with mini or glimmer as judge to bound self-preference; the pipeline already supports coexisting judges per cell
- **Second encoder** — one more index (same chunks, different encoder) separates the encoder/chunking confound behind prose's retrieval win
- **More SME queries** — the realistic slice is 11 questions; this is the single highest-leverage data investment

Next experiments from the original program

- **Field-description ablation** — bare names vs. inline descriptions vs. enums; the literature predicts the largest single gain (Gorilla, ToolAlpaca: +30%)
- **Direct comprehension** — deterministic Q&A over single records, no retrieval, no judge
- **Sparse-metadata stress test** — ablate fields, measure hallucination vs. honest gap-reporting per representation
- **Agentic/MCP arm** — static representation vs. tool-calling agent (nasa/earthdata-mcp already exists)

Takeaways for CMR

1. **Publish a curated projection as a first-class AI representation.** A 32-field facet of UMM was more correct *and* 60% cheaper than the raw record for every model tested. This is the actionable finding, and it is not close.
2. **Ship a Metadata-as-Text summary alongside it.** Prose won retrieval in every condition, is the cheapest rendering to embed, and stayed top-two on reasoning — the strongest single default for embedding pipelines.
3. **Keep the canonical UMM record untouched.** The projection is an *added* representation; precision, provenance, and audit stay with the record of authority.
4. **Do not assume JSON-LD helps LLM consumption.** It was the most expensive rendering and the worst retriever here. Its value is interoperability and linking — serve it for that, not as LLM context.
5. **Content curation first, format second.** Deciding *which fields* go into context beats any serialization debate by a wide margin.
6. **Fund the harness, not a format.** Winners are model- and encoder-dependent and models change; a repeatable, audited evaluation pipeline is the durable artifact. (This one is open and resumable.)

Measure, don't assume.

The best representation is not universal — and now that's a measurement, not a slogan.

Pipeline & analysis	ai-metadata repo — reports/STAGED_PIPELINE.md, reports/ANALYSIS_
Retrieval study	reports/CHUNKED_RETRIEVAL_511Q.md
Literature synthesis	reports/Report.md
Contact	Bernard Benson